Bayesian Correlation Filter Learning With Gaussian Scale Mixture Model for Visual Tracking

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***Abstract*— Correlation filters (CF), a popular tool for visual tracking, suffer from unwanted boundary effects due to the periodic assumption needed for FFT implementation. To address this issue, spatially regularized discriminative correlation filters (SRDCF) have been proposed by introducing a weighting matrix to the regularization term. However, the existing design of spatial weighting matrix is often heuristic and non-adaptive. Inspired by recent advances in joint discrimination and reliability learning for correlation tracking, we propose a principled Bayesian correlation filter learning method using Gaussian scale mixture (GSM) model. The key idea is to decompose each CF coefficient into the product of a positive scalar multiplier and a Gaussian random variable. Treating positive multipliers as weighting coefficients, GSM-based modeling of CFs leads to a spatially adaptive regularization strategy with improved capability of handling various appearance-related uncertainty factors (e.g., scale variation, out-of-plane rotation, and motion blur). Moreover, by imposing a sparse prior over the multipliers, we can jointly learn multipliers and CFs under a unified Bayesian estimation framework. Structured GSM model allows us to better exploit the spatial correlations among CFs and further improve the tracking performance. Experimental results on OTB-2013, OTB-2015, Temple Color-128, VOT-2016, and VOT-2017 show that our tracking method performs favorably when compared with current state-of-the-art methods.**

***Index Terms*— Correlation filters (CF), Gaussian scale mixture, Bayesian tracking, spatially adaptive regularization.**

I. Introduction

VISUAL tracking [1], [2], [4]–[6] is a widely studied problem in computer vision with many important applications in our daily lives (e.g., intelligent surveillance and autonomous driving). Since the available training samples are always limited, it is often challenging to design a robust tracking technique to handle various uncertainty factors such as motion blur/deformation, partial/total occlusion, scale changes, and background clutter [18]. In recent years, correlation filters (CFs) based trackers [3], [12]–[14], [23], [25], [53] have drawn increasingly more attention due to their excellent performance and low computational complexity. The CFs are learned by minimizing a Least-Squares loss on a set of circularly shifted training samples. Along this line of research, the performance of CF trackers has significantly improved e.g., multi-dimensional features [34], [45], [69], multi-kernel method [55], scale estimation [15], [36], [37], [43], [54], nonlinear kernels [44], structural constraints [11], [56], [61] and improved learning methods [27], [57]–[60]. The use of deep convolutional neural network (DCNN) features has further boosted the performance of CF trackers [10], [22], [24], [26], [31], [62], leading to state-of-the-art results on recent benchmarks.

Despite the effectiveness of CF trackers, they still suffer from boundary effects due to the periodic assumption of training samples. The circularly shifted samples acting as negative samples induce notorious boundary effects and reduce the learning accuracy. To address this issue, a dense sampling method for CF learning has been proposed in [57]. The other way to reduce the boundary effects is to introduce a spatial regularization for CF based trackers (SRDCF) [21] by penalizing the filters with a weighting matrix. The boundary effects can also be effectively suppressed by using real negative examples extracted from the background in background-aware CF (BACF) [58]. Meanwhile, efficient convolution operators (ECO) [24] enable a speed-up tracker by compressing feature dimensions and less frequently updating the learned filters. However, in terms of spatial regularization, ECO still employs a fixed handcrafted matrix during the tracking process like in SRDCF [21], which cannot characterize the appearance variation of multiple objects at different times. More recently, a ℓ_1-sparsity regularization has been proposed in [62], leading to a substantial improvement in visual tracking performance. However, the regularization parameter controlling the sparsity of DCF has to be carefully selected. Along this line of research, a novel CF learning method has been proposed in [26] by considering both the discrimination and reliability measures, which has significantly enhanced the performance in terms of suppressing the boundary effect of CF. In [91], a tracking-by-detection model uses sequential detection and group behavior model to solve the problem of incorrect estimation when severe occlusion occurs between targets.

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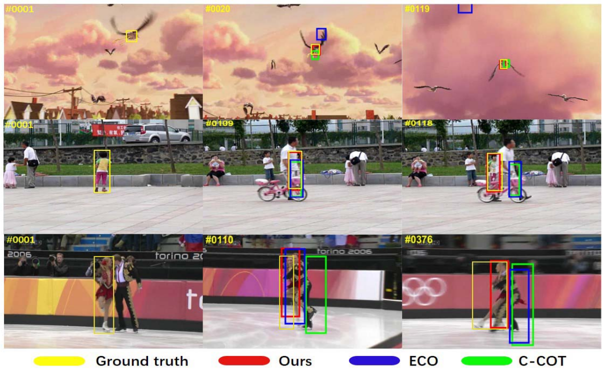

Fig. 1. Tracking result comparison between this work and two recent CF-based methods on OTB benchmark. Both C-COT [25] and ECO [24] using non-adaptive spatial regularization terms suffer from poor tracking performance in cases of short-term out-of-view (top row), occlusion (middle row) and motion blur (bottom row). Our method using spatially adaptive regularization achieves much better performance.

Based on the above observations, we propose to use structured sparsity models to characterize spatial correlation and distinguish the target of interest from the cluttered background via patch discrimination and reliability learning in this paper. More specifically, we propose a Bayesian correlation filter (BCF) learning method with Gaussian scale mixture (GSM) model [71] to adaptively regularize the CF. The key idea is to model each CF coefficient as the product of a hidden positive multiplier and a Gaussian random variable characterizing spatially varying and invariant components respectively. The hidden multipliers served as adaptive regularization coefficients can be jointly estimated along with the Gaussian random variable coefficients under a Bayesian estimation framework, which demonstrates the improved modeling capability for suppressing the boundary effect. Moreover, to exploit the spatial correlations between similar CF coefficients, a structured extension of GSM model has been developed for further improvements in challenging tracking situations (e.g., out-of-view, occlusion and motion blur as shown in Fig. 1). Experimental results on OTB-2013, OTB-2015, Temple Color-128, VOT-2016, and VOT-2017 show that the proposed method performs well among the state-of-the-art algorithms.

The key contributions of this paper are summarized as:

- A BCF learning method based on the GSM model is proposed. The hidden variables used for adaptively regularizing the CF are jointly estimated along with the CFs under the maximum a posterior (MAP) estimation framework. To our knowledge, this is the first attempt to learn the CFs from the Bayesian estimation point of view, which extends the previous work [26].
- Structured extension of GSM model [16] allows us to make a better use of the spatial correlation between CF coefficients to further improve the tracking performance.
- Similar to [57], the optimization problem can be solved by an alternating direction method of multipliers of (ADMM)-based method. Experimental results show that our algorithm performs favorably when compared with current state-of-the-art trackers.

II. Related Works

In this section, we review CF methods associated with visual tracking and GSM modeling techniques for spatial adaptation.

A. Correlation Filter-Based Tracking

Due to its excellent tracking performance, correlation filters (CFs) have attracted wide attentions in recent years. Based on the periodic assumption of the training samples, the learning of CFs admits an efficient implemented via FFT. The MOSSE [28] tracker is first introduced using single channel samples and extended for multichannel feature representation (e.g., HOG [50], Color-Name [51]) leading to significant performance improvements. In [44], the kernel trick has also been employed to learn kernelized CFs. Due to the excellent performance of deep convolutional neural networks (DCNN) in other tasks, the DCNN features have also been used to learn the CFs. Features extracted from the deep network of classification tasks are weak in distinguishing the dynamic changes of targets and background. To help the tracker deal with abrupt appearance variations of the target, dynamic appearance information was extracted from target whose appearance changes rapidly between consecutive frames in [7]. In [8], a bidirectional incongruity-aware model was proposed to make full use of interframe information to improve the performance of the tracker, the generalization ability of the tracker for the subsequent frames was enhanced. In [10], a two-stage positioning method combining *Siamfc* [68] and *staple* [69] was adopted to improve the overall performance, and the proposed model switch can be used in other similar frameworks. A tracking model based on sparse and low-rank was proposed in [11], with the constraints of spatial-temporal-channel, the tracker can adaptively enhance the interpretability and discrimination. In [12], an adaptive discriminative deep correlation filter was proposed by combining adaptive appearance modeling with discriminative feature fine-tuning, which alleviated the problem of drift. In [13], a method of fusing image feature domain and kernel feature domain was proposed to reduce the gap between cyclic filtering and classical filtering algorithms, which improved the recognition ability of learning filters. The scale estimation methods [15], [37] and structured CFs [56] have also been proposed to boost the performance of CFs.

B. Spatially Regularized Correlation Filters

As implicitly circularly shifted training samples induce spatial boundary effects, they inevitably degrade the performance of CF trackers. In [14], channel reliability is taken into account to avoid the performance degradation arising from unreliable/corrupted channels. Through adaptive weight allocation, the influence of reliable channels is increased, and the weight of corrupted channels is reduced, which makes the tracker more discriminative. In [9], a channel graph regularization model was used to address the problem of assigning different weights to similar feature channels, the features that are unfavorable to tracking are effectively suppressed. In [57], this issue was partially resolved by learning CFs with samples containing fewer boundary effects, and the constrained optimization problem was solved by ADMM. Unfortunately, only single-channel training samples are considered in the model. The BACF [58] tracker, as an extension of [57] from single-channel to multichannel, has also been proposed to learn the filters from real background patches to exclude the boundary effects. Different from [57], [58], SRDCF [21] and its variations [24], [25] proposed the spatially regularized DCFs by introducing a weighted regularization term to penalize the fixed CF coefficients in the boundary regions. In [62], the sparsity constraint has been proposed to adaptively select the spatial features for CFs learning. However, the regularization parameters controlling the sparsity of CFs have to be manually tuned. In [84], a regularization term and a mask matrix are jointly learned to express the aberrant term and expand the search region, respectively. The ASRCF [77] exploits an adaptive spatial regularization to alleviate the boundary effects. By learning an effective spatial weight for a specific target, the track can obtain more reliable filter coefficients. In [85] and [86], the spatial weight is integrated with the target saliency information to boost the accuracy of the tracker. In [87], a decision policy is constructed in which spatial weights should be selected to locate the target during the tracking process. In [89], an object-adaptive spatially regularized CF model was developed, different from [77], it learned the object information within a data-driven manner and led to better tracking performance.

C. Gaussian Scale Mixture Modeling

As an effective image modeling tool, the Gaussian scale mixture (GSM) models [63], [70] have shown impressive performance for image restoration [16], [71]. In [16], the sparse coding coefficients were characterized by a GSM model in which both the multipliers and sparse coefficients are jointly estimated. In [71], wavelet coefficients were modeled by GSM as a product of hidden positive scalar multipliers and Gaussian vectors. In [72], the GSM model was extended into the field-of-GSM models by the product of independent homogeneous Gaussian Markov random fields. In [65], the foreground target is decomposed by the structural GSM model, which facilitates the separation of the foreground target from the cluttered background. Built upon [64], [65] exploits temporal dependency, which further improved the reconstruction accuracy of foreground targets. In [67], mixed noise coefficients are characterized by a Laplacian scale mixture model, which achieves the good performance for multi-frame image denoising. This model has also achieved excellent results in the denoising of video data, which demonstrates the robustness of the model in temporal domain [66]. Unlike previous works, we propose to use GSM to model the CFs coefficients in this paper. Specifically, the positive scalar multipliers are used to model the parts of the targets that are beneficial for tracking. With a sparse hyper prior, we are able to adaptively suppress the boundary effects and the cluttered background by jointly estimating the hidden variable and the CFs.

III. Bayesian Correlation Filter Learning With Gaussian Scale Mixture Model

In this section, we first briefly review the conventional SRDCF learning method and then introduce the proposed Bayesian CF (BCF) learning method.

A. Conventional SRDCF Learning

Let $\mathcal{D} = \{(\mathbf{x}_t, \boldsymbol{y}_t)\}_{t=1}^{T}$ denote a set of training images, where $\mathbf{x}_t = [\boldsymbol{x}_{t,1}, \ldots, \boldsymbol{x}_{t,K}] \in \mathbb{R}^{N\times K}$ contains K vectorized feature maps with size of N, $\boldsymbol{y}_t \in \mathbb{R}^N$ denotes the predefined Gaussian shape labels. The function of SRDCF learning can be formulated as

$$\min_{\mathbf{h}} \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k||_2^2 + \lambda \sum_{k=1}^{K} ||\mathbf{w}\boldsymbol{h}_k||_2^2, \quad (1)$$

where $*$ denotes the circular convolution operator, $\mathbf{h} = [\boldsymbol{h}_1, \ldots, \boldsymbol{h}_K] \in \mathbb{R}^{N\times K}$ is the CF, $\mathbf{w} \in \mathbb{R}^{N\times N}$ is a predefined matrix for spatial regularization, and π_t is a weight assigned to the t-th sample. Note that $\mathbf{w}$ is spatially invariant, it is natural to consider a spatially adaptive extension to better characterize appearance variation (e.g., out-of-view, occlusion, and motion blur) in video.

B. Correlation Filter Learning via GSM Model

One plausible attack is to extend the CF learning problem from the Bayesian point of view. Let $\boldsymbol{y}_t = \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k + \boldsymbol{e}_t$, $k = 1, \ldots, K$, where $\boldsymbol{e}_t \sim N(\mathbf{0}, \sigma_t^2 \mathbf{I})$ denotes the approximation error of the t-th sample and σ_t^2 denotes the variance of Gaussian random variable $\boldsymbol{e}_t$. Then CF can be learned by maximizing the following posterior probability

$$\begin{aligned}\log p(\mathbf{h}|\mathcal{D}) &\propto \log p(\mathcal{D}|\mathbf{h}) + \log p(\mathbf{h}) \\ &= \sum_{t=1}^{T} \log p(\boldsymbol{y}_t|\mathbf{x}_t, \mathbf{h}) + \log p(\mathbf{h}),\end{aligned} \quad (2)$$

where $p(\mathbf{h})$ and $p(\boldsymbol{y}_t|\mathbf{x}_t, \mathbf{h})$ denote the prior distribution of $\mathbf{h}$ and the Gaussian likelihood term, respectively. The likelihood term is given by

$$p(\boldsymbol{y}_t|\mathbf{x}_t, \mathbf{h}) \propto \frac{1}{\sigma_t^2} \exp(-\frac{1}{2\sigma_t^2} ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k||_2^2). \quad (3)$$

If each CF is modeled by Gaussian, $p(\mathbf{h})$ can be written as

$$p(\mathbf{h}) = \prod_{k=1}^{K} p(\boldsymbol{h}_k) = \prod_{k=1}^{K} \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}\theta_{k,i}} \exp(-\frac{h_{k,i}^2}{2\theta_{k,i}^2}), \quad (4)$$

where $\theta_{k,i}$ denotes the variance of $h_{k,i}$. By substituting Eqs. (3) and (4) for the MAP estimation of Eq. (2), it can be seen that the objective function in our BCF plays an equivalent role to that of Eq. (1) in SRDCF when we set $\sigma_t^2 = \frac{\lambda}{\pi_t}$ and $\mathbf{w}_k = \mathrm{diag}(\frac{1}{\theta_{k,i}^2 + \epsilon})$. If all CFs of different channels share the same prior, the spatially adaptive BCF degenerates into the special case of SRDCF. Note that since the variance parameters $\theta_{k,i}$ depend on the video content, it is often challenging to manually tune them on a pixel-by-pixel basis.

To solve this problem, we model the CFs by the GSM model that has shown effective for spatially adaptive image restoration [16]. Specifically, we can decompose each $\boldsymbol{h}_k$ as the point-wise product of a hidden positive multiplier vector $\boldsymbol{\theta}_k$ and a Gaussian vector $\boldsymbol{\alpha}_k$, i.e., $h_{k,i} = \theta_{k,i}\alpha_{k,i}$. Conditioned on $\theta_{k,i}$, $\alpha_{k,i}$ can be regarded as Gaussian function with a standard deviation $\theta_{k,i}$. Assuming that $\theta_{k,i}$ are i.i.d and independent of $\alpha_{k,i}$, the GSM modeling of $\boldsymbol{h}_k$ can be defined by

$$p(\boldsymbol{h}_k) = \prod_{i=1}^{N} p(h_{k,i}),\ p(h_{k,i}) = \int_0^{\infty} p(h_{k,i}|\theta_{k,i})p(\theta_{k,i})d\theta_{k,i}, \tag{5}$$

where $p(h_{k,i}|\theta_{k,i})$ is the conditional distribution function with known variance parameter $\theta_{k,i}$ and $p(\theta_{k,i})$ represents the prior distribution of $\theta_{k,i}$. Generally speaking, it is difficult to directly optimize the MAP estimation of $\boldsymbol{h}_k$ using the GSM model, because there is no analytical solution to $p(\boldsymbol{h}_k)$. However, such difficulty can be overcome by jointly estimating $\boldsymbol{h}_k$ and $\boldsymbol{\theta}_k$ as following

$$\min_{\Theta,\mathbf{h}} \sum_{t=1}^{T} \log p(\boldsymbol{y}_t|\mathbf{x}_t, \mathbf{h}) + \sum_{k=1}^{K} p(\boldsymbol{h}_k|\boldsymbol{\theta}_k) + \sum_{k=1}^{K} p(\boldsymbol{\theta}_k), \tag{6}$$

where $\Theta = [\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K]$ denotes the positive multiplier matrix. For the prior $p(\boldsymbol{\theta}_k)$, we use the noninformative prior-i.e., the Jeffrey's prior [73] $p(\boldsymbol{\theta}_k) = \prod_i p(\theta_{k,i}) = \prod_i \frac{1}{\theta_{k,i}+\epsilon}$. By substituting $p(\boldsymbol{h}_k|\boldsymbol{\theta}_k)$, $p(\boldsymbol{\theta}_k)$ and the Gaussian likelihood term into Eq. (6), the objective function can be rewritten as follows

$$(\mathbf{h}, \Theta) = \arg\min_{\mathbf{h},\Theta} \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k||_2^2 + \lambda \sum_{k=1}^{K} ||\mathbf{w}_k \boldsymbol{h}_k||_2^2 + 4\lambda \sum_{k=1}^{K} \log(\boldsymbol{\theta}_k + \epsilon), \tag{7}$$

when compared with Eq. (1) where the regularization matrix $\mathbf{w}$ has to be predefined, both CFs and weighting matrix are jointly estimated in the new objective function of Eq. (7). Note that in the GSM modeling, $\boldsymbol{h}_k$ can be written as $\boldsymbol{h}_k = \Lambda_k \boldsymbol{\alpha}_k$, where $\Lambda_k = \mathrm{diag}(\theta_{k,i}) \in \mathbb{R}^{N\times N}$. It follows that the above objective function can be rewritten into

$$(\mathbf{A}, \Theta) = \arg\min_{\mathbf{A},\Theta} \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * (\Lambda_k \boldsymbol{\alpha}_k)||_2^2 + \lambda \sum_{k=1}^{K} ||\boldsymbol{\alpha}_k||_2^2 + 4\lambda \sum_{k=1}^{K} \log(\boldsymbol{\theta}_k + \epsilon), \tag{8}$$

where $\mathbf{A} = [\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_k]$ denotes the GSM multiplier matrix. With the sparse prior, $\boldsymbol{\theta}_k$ can adaptively characterize salient regions for visual tracking and suppress boundary effects as well as background clutter.

As shown in Fig. 2, the spatial regularization $\mathbf{w}$ for the classical algorithm SRDCF has an inverse Gaussian shape, which does not vary during the tracking process. By contrast, under the maximum a posterior (MAP) estimation framework, GSM decomposition $\boldsymbol{h} = \boldsymbol{\theta} * \boldsymbol{\alpha}$ allows us to adaptively capture the nonstationarity in video (e.g., due to occlusion) by varying the shape of the parameter $\boldsymbol{\theta}$. When the woman's face is occluded by a book, the spatial distribution of $\boldsymbol{\theta}$ values can evolve with frame numbers, which achieves an improved tradeoff between the reliability (as highlighted by the yellow color) and the discrimination from the background

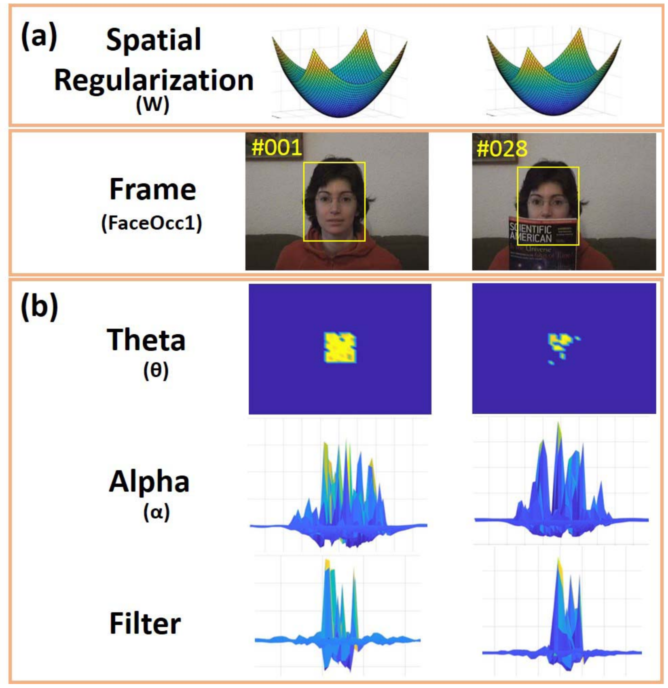

Fig. 2. An visualization of SRDCF (a) and BCF (b) for video sequence 'FaceOcc1'. SRDCF uses a fixed spatial regularization kernel; by contrast, the proposed method can learn a flexible time-varying sparse matrix to alleviate boundary effects. Our method can better tolerate appearance variations of the target implying improved robustness.

(as highlighted by the blue color) [26]. Note that this sparsity mask is used to select which parts are more useful for $\boldsymbol{\alpha}$, the bottom row (labeled by "Filter") of Fig. 2 denotes the filters (h) learned through a series of optimization steps.

C. Structured GSM for Correlation Filter Learning

Another important new insight brought by this work is the exploitation of spatial correlation in addition to spatial adaptation. In Eq. (8), the CF coefficients are assumed to be i.i.d. However, a key observation is that the neighboring CF coefficients are often spatially correlated, e.g., the CF coefficients corresponding to the same object (e.g., the moving target, the background) are highly correlated. To exploit the spatial correlations, we extend the GSM model into a structured GSM model by characterizing the neighboring CF coefficients with the same GSM model. Assuming that the filter coefficients are grouped into S groups, $\mathcal{G}_s$, $s = 1, \ldots, S$, the structured GSM model can be written as

$$p(\boldsymbol{h}_k) = \prod_{s=1}^{S} \prod_{j\in\mathcal{G}_s} p(h_{k,j}),$$
$$p(h_{k,j}) = \int_0^{\infty} p(h_{k,j}|\theta_{k,s})p(\theta_{k,s})d\theta_{k,s},\ j \in \mathcal{G}_s. \tag{9}$$

In theory, it is desirable to cluster the coefficients corresponding to the same object or background regions into the same group. Considering the strong correlation between adjacent pixels, we divide the filter coefficients into several non-overlapped rectangle blocks of size $r \times c$, and assume that the coefficients within the same block share the same prior.

By substituting the structured GSM model into Eq. (9) in the MAP estimation of Eq. (2), a structured GSM model-based CFs learning method can be obtained as follows

$$(\mathbf{A}, \Theta) = \arg\min_{\mathbf{A},\Theta} \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * (\Lambda_k \boldsymbol{\alpha}_k)||_2^2 + \lambda \sum_{k=1}^{K} ||\boldsymbol{\alpha}_k||_2^2 + 4\lambda \sum_{k=1}^{K}\sum_{s=1}^{S} |\mathcal{G}_s| \log(\theta_{k,s} + \epsilon), \quad (10)$$

where $\Lambda_k = \mathrm{diag}(\theta_{k,s_i}) \in \mathbb{R}^{N\times N}$, $s_i \in \{1,\ldots,S\}$ denotes the group where the i-th coefficient belongs to, $\theta_{k,s}$ is the positive multiplier for the s-th group of k-th channel and $|\mathcal{G}_s|$ denotes the number of coefficients of the s-th group. As shown in Fig. 3, the top row represents the target status at different timing, and the yellow box denotes the target that needs to be tracked. We use the parameters of $\boldsymbol{\theta}$ learned at three different frames to illustrate the difference between the original GSM (middle row) and Structured GSM (bottom row). The yellow parts represent the learned effective parameters, and the blue parts refer to the background clutter information. We can see that the $\boldsymbol{\theta}$ of original GSM is more scattered and sparse than that of the Structured GSM. By taking the structured information within the target into consideration, the learned parameters of $\boldsymbol{\theta}$ appear to be more clustered and continuous.

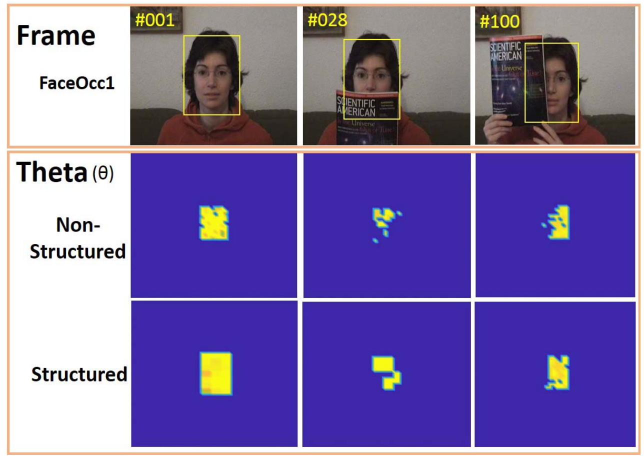

Fig. 3. Comparison of positive scaling variable ($\boldsymbol{\theta}$) with GSM (middle row) and Structured GSM (bottom row) at different times. The distribution of $\boldsymbol{\theta}$ parameters in GSM are more random and has less spatial correlation among the CF coefficients than Structured GSM.

IV. Optimization Algorithm

The objective function of Eq. (10) can be solved by using the ADMM [57]. By introducing an auxiliary variable $\boldsymbol{h}_k$, we can obtain the following augmented Lagrangian function as

$$\mathcal{L}(\mathbf{h}, \Lambda, \boldsymbol{\alpha}, \boldsymbol{\eta}) = \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k||_2^2 + \gamma \sum_{k=1}^{K} ||\boldsymbol{h}_k - \Lambda_k \boldsymbol{\alpha}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}||_2^2 + \lambda \sum_{k=1}^{K} ||\boldsymbol{\alpha}_k||_2^2 + 4\lambda \sum_{k=1}^{K}\sum_{s=1}^{S} |\mathcal{G}_s| \log(\theta_{k,s} + \epsilon), \quad (11)$$

where $\boldsymbol{\eta}_k$ is the Lagrangian multiplier and $\gamma > 0$. Minimizing the augmented Lagrangian function equals to alternatively solving the following subproblems,

$$\min_{\boldsymbol{h}_k} \sum_{t=1}^{T} \pi_t ||\boldsymbol{y}_t - \sum_{k=1}^{K} \boldsymbol{x}_{t,k} * \boldsymbol{h}_k||_2^2 + \gamma \sum_{k=1}^{K} ||\boldsymbol{h}_k - \Lambda_k \boldsymbol{\alpha}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}||_2^2, \quad (12)$$

$$\min_{\Lambda_k} \gamma \sum_{k=1}^{K} ||\boldsymbol{h}_k - \Lambda_k \boldsymbol{\alpha}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}||_2^2 + 4\lambda \sum_{k=1}^{K}\sum_{s=1}^{S} |\mathcal{G}_s| \log(\theta_{k,s} + \epsilon), \quad (13)$$

$$\min_{\boldsymbol{\alpha}_k} \gamma \sum_{k=1}^{K} ||\boldsymbol{h}_k - \Lambda_k \boldsymbol{\alpha}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}||_2^2 + \lambda \sum_{k=1}^{K} ||\boldsymbol{\alpha}_k||_2^2, \quad (14)$$

$$\boldsymbol{\eta}_k^{(l+1)} = \boldsymbol{\eta}_k^{(l)} + \gamma (\boldsymbol{h}_k^{(l+1)} - \Lambda_k^{(l+1)} \boldsymbol{\alpha}_k^{(k+1)}), \quad (15)$$

where l denotes the number of iterations. The detailed solutions of each subproblem are given as follows.

1) The h-Subproblem: Using the Parseval's theorem, the $\mathbf{h}$-subproblem can be formulated in the Fourier domain as

$$\min_{\hat{\boldsymbol{h}}_k} \sum_{t=1}^{T} \pi_t ||\hat{\boldsymbol{y}}_t - \sum_{k=1}^{K} \hat{\boldsymbol{x}}_{t,k} \odot \hat{\boldsymbol{h}}_k||_2^2 + \gamma \sum_{k=1}^{K} ||\hat{\boldsymbol{h}}_k - \hat{\boldsymbol{g}}_k + \frac{\hat{\boldsymbol{\eta}}_k}{2\gamma}||_2^2, \quad (16)$$

where $\boldsymbol{g}_k = \Lambda_k \boldsymbol{\alpha}_k$, and $\hat{\boldsymbol{h}}_k$, $\hat{\boldsymbol{g}}_k$ and $\hat{\boldsymbol{x}}_{t,k}$ are the discrete Fourier representations of $\boldsymbol{h}_k$, $\boldsymbol{g}_k$ and $\boldsymbol{x}_{t,k}$, respectively, and $\odot$ denotes the point-wise product. The closed-form solution for Eq. (16) can be obtained as [24]

$$\hat{\boldsymbol{h}}_k = (\mathbf{X}^H \Gamma \mathbf{X} + \gamma \mathbf{I})^{-1} (\mathbf{X}^H \Gamma \hat{\boldsymbol{y}} + \gamma \hat{\boldsymbol{g}}_k - \hat{\boldsymbol{\eta}}_k), \quad (17)$$

where $\mathbf{I}$ is an identity matrix, Γ is a diagonal matrix of the weights π_t, and $\mathbf{X}$ is a sparse matrix with diagonal blocks consisting of elements from $\hat{\boldsymbol{x}}_{t,k}$. Eq. (17) can be computed using the Conjugate Gradient (CG) method. Finally, $\boldsymbol{h}_k$ is obtained from $\hat{\boldsymbol{h}}_k$ using the inverse DFT.

2) The $\boldsymbol{\theta}$-Subproblem: Each $\boldsymbol{\theta}_k$ can be solved by minimizing

$$\min_{\boldsymbol{\theta}_k} ||\boldsymbol{h}_k - \Lambda_k \boldsymbol{\alpha}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}||_2^2 + \frac{4\lambda}{\gamma} \sum_{s=1}^{S} |\mathcal{G}_s| \log(\theta_{k,s} + \epsilon), \quad (18)$$

since each $\theta_{k,s}$ is independent, the above objective function can be solved by a series of scalar optimization methods

$$\theta_{k,s} = \arg\min_{\theta_{k,s}} \sum_{j \in \mathcal{G}_s} (h_{k,j} - \theta_{k,s}\alpha_{k,j} + \frac{\eta_{k,j}}{\gamma})^2 + \frac{4\lambda|\mathcal{G}_s|}{\gamma} \log(\theta_{k,s} + \epsilon), \ \ s.t. \ \theta_{k,s} \geq 0, \quad (19)$$

which can be further rewritten as

$$\theta_{k,s} = \arg\min_{\theta_{k,s}} a\theta_{k,s}^2 + b\theta_{k,s} + c \log(\theta_{k,s} + \epsilon), \quad (20)$$

where $a = \sum_{j\in\mathcal{G}_s} \alpha_{k,j}^2$, $b = -2\sum_{j\in\mathcal{G}_s} \alpha_{k,j}(h_{k,j} + \frac{\eta_{k,j}}{\gamma})$, and $c = \frac{4\lambda|\mathcal{G}_s|}{\gamma}$. Let $f(\theta_{k,s})$ denote the right-hand side of Eq. (20). By taking $\frac{df(\theta_{k,s})}{d\theta_{k,s}} = 0$, a closed-form solution can be obtained as

$$\theta_{k,s} = \begin{cases} 0, & \text{if } b^2 - 4ac < 0 \\ \arg\min_{\theta_{k,s}}\{f(0), f(\theta^*)\}, & \text{otherwise} \end{cases}, \tag{21}$$

where θ^* is given by

$$\theta^* = \frac{-(b+a\epsilon) \pm \sqrt{(b+a\epsilon)^2 - 4a(c+b\epsilon)}}{2a}. \tag{22}$$

3) The $\boldsymbol{\alpha}$-Subproblem: For fixed $\boldsymbol{h}_k$, $\boldsymbol{\theta}_k$ and $\boldsymbol{\eta}_k$, $\boldsymbol{\alpha}_k$ can be solved by minimizing

$$\boldsymbol{\alpha}_k = \arg\min_{\boldsymbol{\alpha}_k} \gamma\, ||\boldsymbol{h}_k + \frac{\boldsymbol{\eta}_k}{2\gamma} - \Lambda_k \boldsymbol{\alpha}_k||_2^2 + \lambda||\boldsymbol{\alpha}_k||_2^2, \tag{23}$$

the solution is as follows

$$\boldsymbol{\alpha}_k = (\Lambda_k^\top \Lambda_k + \frac{\lambda}{\gamma}\mathbf{I})^{-1}\Lambda_k^\top(\boldsymbol{h}_k + \frac{\boldsymbol{\eta}_k}{2\gamma}), \tag{24}$$

where Λ_k represents a diagonal matrix.

4) The Update of $\boldsymbol{\eta}_k$ and γ: The Lagrangian multiplier $\boldsymbol{\eta}_k$ is iteratively updated as in Eq. (15). The parameter γ is updated by

$$\gamma^{(l+1)} = \min(\rho\gamma^{(l)}, \gamma_{\max}), \tag{25}$$

where $\rho > 1$ is a pre-defined scalar controlling the convergence speed and γ_{max} denotes the maximum value of γ.

V. Experimental Results

In this section, we perform a comprehensive evaluation of our Bayesian correlation filter (BCF) model by comparing with the state-art-the-art trackers on five benchmark datasets including OTB-2013 [17], OTB-2015 [18], Temple Color-128 [74], VOT-2016 [19], and VOT-2017 [75].

A. Experimental Setups

1) Implementation Details: In visual tracking [48], [49], robust features can improve visual tracking performance. Therefore, as in most of the existing visual trackers [22]–[25], we also exploit standard hand-crafted features and deep features for fair comparisons. The hand-crafted features include 31-channel HOG [50] and 10-channel Color Names (CN) [51], and the deep features are the output of Conv1 and Conv4-3 from VGG-M network [52] as in [24], [25]. We analyze the effectiveness of the proposed model by using different features in terms of hand-crafted features (HOG, Color Names) and CNN features.

The tracker with only hand-crafted features is denoted as BCF-HC, and the BCF is based on both deep and hand-crafted features. In the proposed model, the scale factor ρ, the initial stepsize parameter γ and the maximum value $\gamma_{\max}$ in Eq. (25) are set to 1.2, 10, and 100, respectively. The regularization parameter σ in Eq. (11) is set to 0.001. The iteration number for the ADMM is set to two. The size of the block for the target object is set as $\varsigma_s = 3 \times 2$ in Eq. (13). Note that for fair comparisons with other trackers, we employ the results provided by the original paper. All comparative experiments are implemented with Matlab 2017b and the MatConvnet toolbox [20] on a computer with Inter Core i7-3700K 3.4GHz CPU and a NVIDIA GTX 1080Ti GPU.

2) Datasets: The proposed tracker is evaluated on five benchmark datasets including OTB-2013 [17], OTB-2015 [18], Temple Color-128 [74], VOT-2016 [19], and VOT-2017 [75], which are the most widely used benchmarks in visual tracking. The OTB-2013 and OTB-2015 benchmarks consist of 51 and 100 videos, respectively, with 11 various challenging factors. The datasets contain color and gray images. The Temple Color-128 dataset [74] consists of 129 color sequences. The VOT-2016 [19] contains 60 videos and all of them are marked with an irregular box, which brings more challenges to the task of visual tracking. As an upgrade of VOT-2016, the VOT-2017 dataset [75] uses more challenging videos to replace some simple ones and the total number of videos remains unchanged.

3) Evaluation Metrics: We use the evaluation metrics provided by the respective benchmark to evaluate the proposed algorithm against the most advanced tracking methods. Following the protocol used in [25], [32], [35], [76], we report the comparison results of one-pass evaluation (OPE) for OTB-2013, OTB-2015, and Temple Color-128 datasets. Two evaluation indexes are used to compare all algorithms including success plot and precision plot. (1) The success plot illustrates the ratios of successful frames over the range of thresholds [0, 1], where successful frames are those that overlap more than a given threshold. The area-under-the-curve (AUC) is used to rank all the trackers in the legend. (2) The precision plot describes the score of average distance precision (DP) for each tracker under a series of thresholds. The DP is defined as the percentage of frames whose estimated location is within the given threshold.

The expected average overlap (EAO), robustness raw value (R), and accuracy raw value (A) are used to measure the performance of the model in VOT-2016 and VOT-2017 datasets. The EAO performs both robustness and accuracy. The robustness raw value (R) represents the times of the target loses or fails in the process of tracking. The accuracy raw value (A) denotes the average overlap ratio between the ground truth and the proposal produced from experiments.

4) Model Update: Most of the existing DCF-based tracking algorithms update their models for each frame, which seriously affects the load of calculation and suffers from the risk of over-fitting. In ECO [24], an infrequent update strategy is used, which not only reduces the amount of computation but also improves the tracking results. We have adopted this new update strategy in our implementation. More detailed information can be found in [24].

B. Overall Performance

1) Experiments on OTB-2013 Dataset:

a) Baseline Methods: We compare the BCF algorithm with 17 related trackers including DSST [15], SRDCF [21], DeepSRDCF [22], ECO [24], C-COT [25], MDNet [29], CREST [32], VITAL [35], SAMF [36],

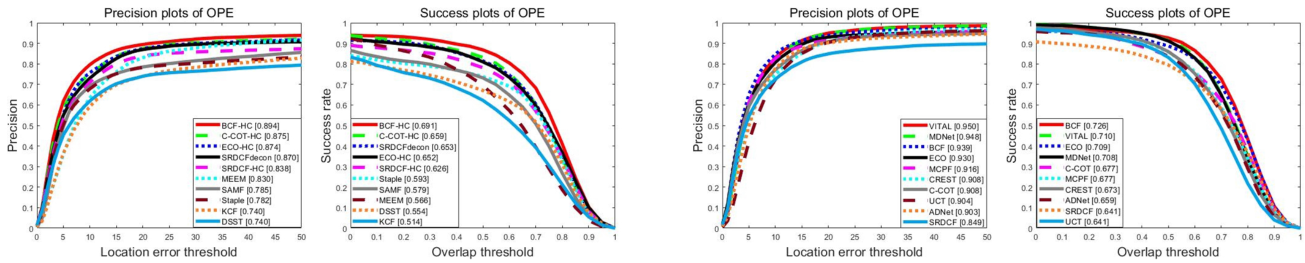

Fig. 4. Success and precision plots on OTB-2013. The BCF model performs more significantly than the related methods. The legend covers AUC and DP scores for all trackers. (a) BCF with hand-crafted features, (b) BCF with deep features.

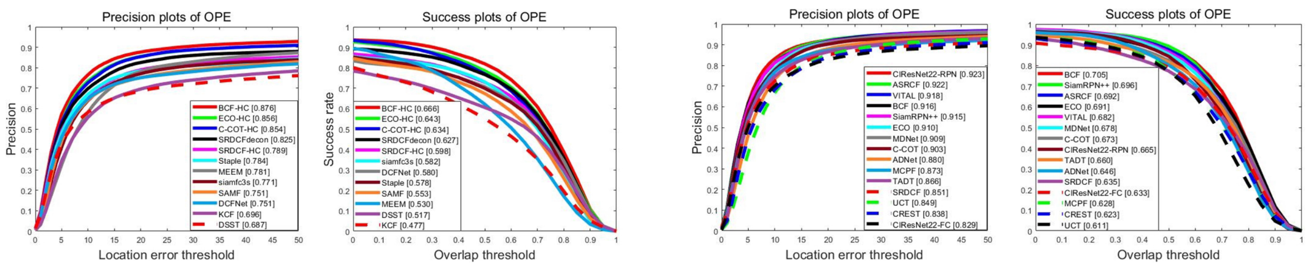

Fig. 5. Success and precision plots using OPE on OTB-2015 benchmark. Both the AUC score and the DP score are listed in the legend. (a) BCF with hand-crafted features, (b) BCF with deep features.

MEEM [38], SRDCFdecon [39], ADNet [41], UCT [42], MCPF [43], KCF [44], RTT [47] and Staple [69].

b) Quantitative Evaluation: We exhibit the comparison results of the top 10 trackers based on hand-crafted and deep features, the success and precision rates all reported in Fig. 4. The proposed BCF approach performs favorably with AUC of (69.1%, 72.6%) and DP of (89.4%, 93.9%) by using hand-crafted and deep features. In particular, based on hand-crafted features, the BCF-HC achieves the best and outperforms the second best method (C-COT-HC) [25] by (4.2%, 1.9%) in terms of AUC and DP score, respectively. The details are shown in Fig. 4(a). SRDCF-HC [21] and SRDCFdecon [39] are two related CF-based trackers, which provide results with a DP score of (83.8%, 87.0%) and an AUC score of (62.6%, 65.3%), respectively. Compared to them, our method gets a significant improvement of (6.5%, 3.8%) in terms of AUC score, and (5.6%, 2.4%) in terms of DP score for the two trackers, respectively.

With hand-crafted and deep features as shown in Fig. 4(b), our BCF algorithm performs superior performance, which provides the results with the AUC and DP score of 72.6% and 93.9%, respectively. In this experimental setting, compared with SRDCF [21], our BCF tracker achieves significant improvement of 8.5% and 9.0% in terms of AUC and DP scores, respectively. In addition, the proposed tracker obtains a relative improvement of (3.1%, 0.9%) and (5.9%, 1.7%) over two state-of-the-art CF-based trackers ECO [24] and C-COT [25] in terms of AUC and DP scores, respectively. When compared with the current topmost trackers including MDNet [29] and VITAL [35], our method achieves comparable performance as (1.8%, 1.6%) in terms of AUC score, respectively. Overall, the scores of DP and AUC demonstrate that the proposed BCF performs well compared with existing state-of-the-art trackers.

2) Experiments on OTB-2015 Dataset:

a) Baseline Methods: We evaluate the performance of our BCF on OTB-2015 dataset with 23 state-of-the-art trackers, including DSST [15], SRDCF [21], DeepSRDCF [22], ECO [24], C-COT [25], MDNet [29], CREST [32], VITAL [35], SAMF [36], MEEM [38], SRDCFdecon [39], DCFNet [40], ADNet [41] UCT [42], MCPF [43], CF2 [45], HDT [46], Siamfc [68], Staple [69], TADT [76], ASRCF [77], SiamRPN++ [82] and CIResNet22 [83].

b) Quantitative Evaluation: We show the success and precision rates of our BCF based on hand-crafted and deep features in Fig. 5. Only the top 12 algorithms with their AUC scores and average distance precision scores are listed in the figure legend. By using hand-crafted features as shown in Fig. 5(a), the proposed BCF-HC model attains the AUC score of 66.6% and DP of 87.6%. In details, BCF-HC tracker also achieves the best by using hand-crafted features and outperforms the second best method (ECO-HC) [24], by 2.3% and 2.0% in terms of AUC and DP metrics, respectively. Compared to SRDCF-HC [21] and SRDCFdecon [39], our tracker obtains a notable gain of (6.8%, 3.9%) in terms of AUC score, and (8.7%, 5.1%) in terms of DP score, respectively. When using hand-crafted and deep features, our BCF tracker can achieve much better performance with the AUC score of 70.5% and DP of 91.6%. The detailed results are shown in Fig. 5(b). In this experimental setting, the proposed tracker gets a relative improvement of (2.7%, 1.4%) and (1.3%, 0.6%) over C-COT [25] and ECO [24] in terms of AUC and DP

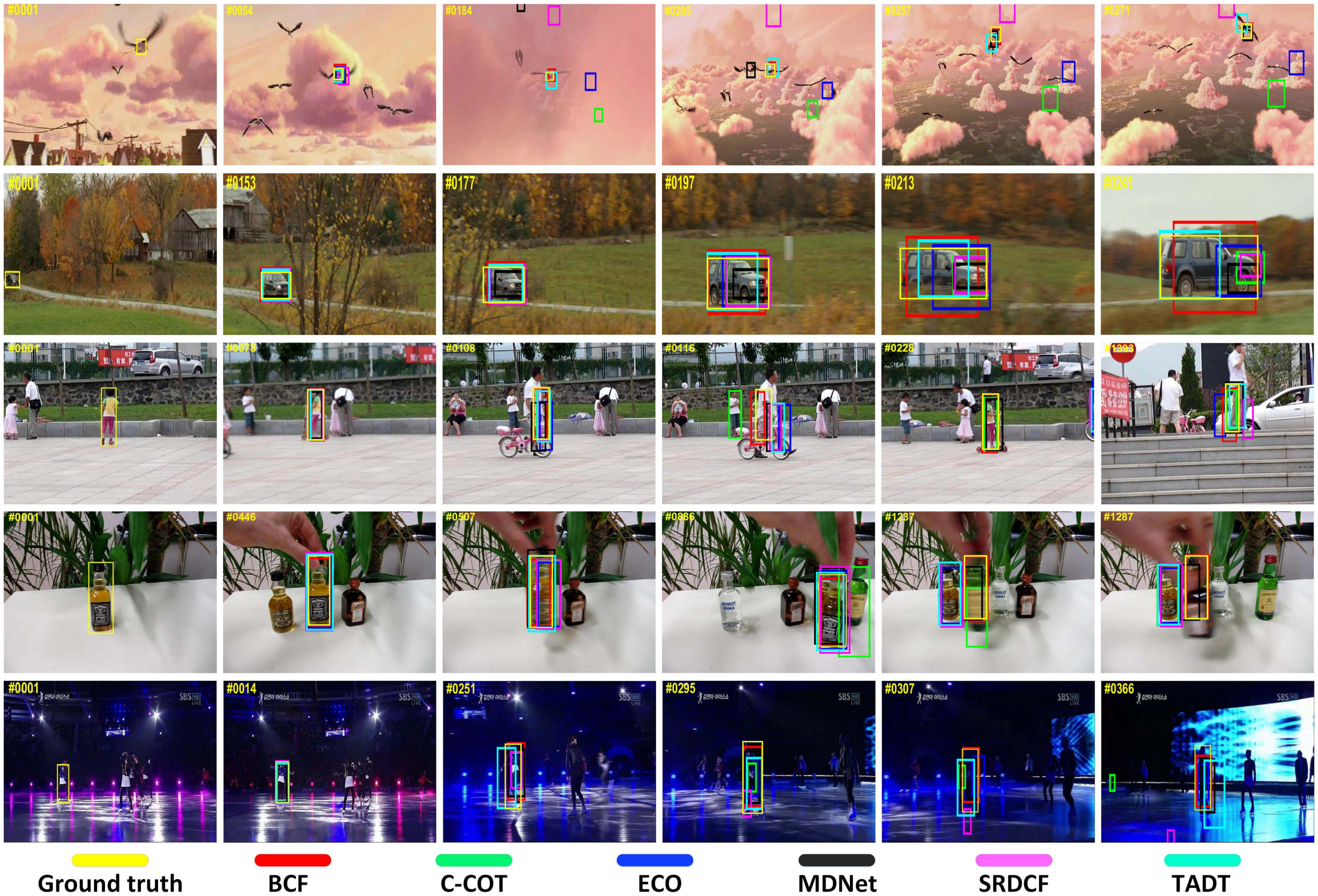

Fig. 6. The results of 6 different trackers on 5 videos, including Bird1, CarScale, Girl2, Liquor and skating1 from the OTB-2015 dataset. It's best to view the details in color.

scores, respectively. And it shows comparable results with the CF-based tracker ASRCF [77]. Compared to deep models such as SiamRPN++ [82] and VITAL [35], our tracker also achieves comparable performance. Overall, the experimental results demonstrate that our BCF model performs favorably among the state-of-the-art trackers.

c) Attribute-based Evaluation: Attribute analysis is helpful to comprehend the performance of the model in different aspects. We analyze the performance of our BCF tracker under 11 challenges [18], e.g., deformation, out-of-plane rotation, scale variation, and occlusion. In Fig. 7, we list 6 attribute plots containing the top 12 trackers. Note that the proposed method achieves promising performance in handling most of the challenges, especially in background clutter, deformation, scale variation, and illumination variation. Compared with existing existing algorithms, our BCF tracker performs well, which demonstrates that our model can select discriminative features for robust visual tracking.

3) Experiments on Temple Color-128 Dataset:

a) Baseline Methods: We evaluate our model in [74] with other trackers including SRDCF [21], DeepSRDCF [22], ECO [24], C-COT [25], SRDCFdecon [39], TADT [76] and ASRCF [77].

b) Attribute-Based Evaluation: We show the comparison results of our BCF based on deep features in Fig. 8. The top 15 trackers with their AUC scores and average distance precision scores are exhibited in the figure legend.

As shown in Fig. 8, the proposed BCF algorithm obtains the AUC score of 61.3% and the DP of 79.8% by using hand-crafted and deep features. Our BCF tracker achieves the best and outperforms the method ECO [24] by 1.6% and 0.1% in terms of AUC and DP metrics, respectively. The proposed tracker gets a relative improvement of 3.9% and 1.7% over C-COT [25] in terms of AUC and DP scores, respectively. Compared to the existing trackers such as SRDCF [21] and DeepSRDCF [22], our BCF obtains an absolute gain of (10.3%, 7.6%) and (10.4%, 6.1%) in terms of AUC and DP scores, respectively. For the topmost trackers as ASRCF [77], our model also achieves comparable performance.

4) Experiments on VOT-2016 Dataset: We evaluate the proposed BCF method on VOT-2016 Dataset [19] with the state-of-the-art trackers including DeepSRDCF [22], STRCF [23], ECO [24], C-COT [25], DRT [26], CSR-DCF [27], MDNet [29], TCNN [30], CREST [32], DaSiamRPN [33], MCCT [34], Staple [69] and SiamRPN [78]. The tracking performance is measured by EAO, A and R [19]. Detailed performance comparison can be seen in Table I. Our

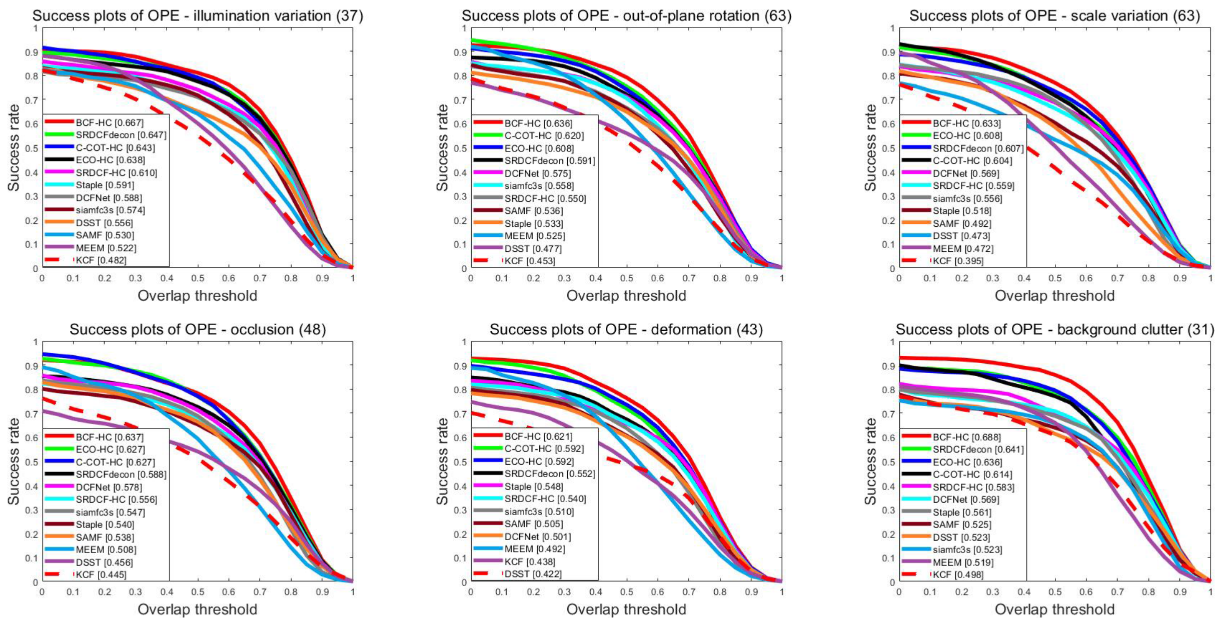

Fig. 7. The assessment based on 6 challenge attributes on OTB-2015 with hand-crafted features. Success plots include illumination variation, out-of-plane rotation, scale variation, occlusion, deformation, and background clutter. Our BCF performs well than other models.

TABLE I

COMPARISON WITH THE RELATED ALGORITHMS IN TERMS OF EXPECTED AVERAGE OVERLAP (EAO), ACCURACY RANK (A), AND ROBUSTNESS RANK (R) ON VOT-2016. TOP THREE INDICATORS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY

	SiamRPN	Staple	MDNet	TCNN	DeepSRDCF	C-COT	ECO	STRCF	CREST	DaSiamRPN	CSR-DCF	MCCT	BCF	DRT
EAO	0.344	0.295	0.257	0.325	0.276	0.331	0.374	0.313	0.283	0.411	0.338	0.393	0.412	0.442
A	0.56	0.54	0.54	0.55	0.53	0.54	0.55	0.55	0.51	0.61	0.51	0.58	0.55	0.57
R	0.26	0.38	0.34	0.27	0.33	0.24	0.20	-	0.24	0.22	-	-	0.18	0.14

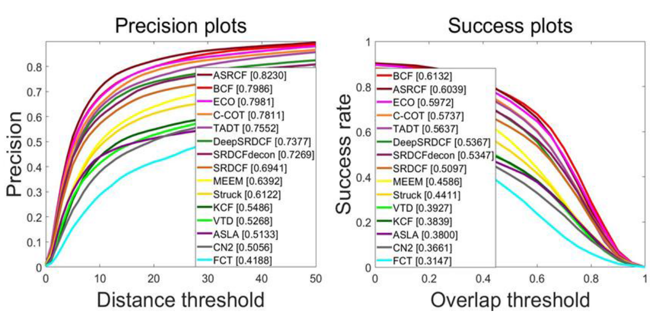

Fig. 8. Performance evaluation of success and precision plots on the Temple Color-128 dataset over all the 129 sequences. Comparison with different trackers based on deep features, the proposed BCF performs favorably.

proposed BCF ranks second with the EAO score of 0.412 compared to the best tracker DRT (0.442). Furthermore, the proposed BCF algorithm outperforms the DeepSRDCF [22] and ECO [24] methods in all measures. Overall, the proposed BCF tracker performs favorably against the state-of-the-art methods.

5) Experiments on VOT-2017 Dataset: We validate the proposed tracker on the VOT-2017 [75]. The comparison results with some trackers are shown in Table II, including ECO [24], C-COT [25], CSR-DCF [27], MCCT [34], MCPF [43], SiamRPN [78], SPM [79], SA-Siam [80], GCT [81], CFCF [49], and some state-of-the-art methods in the VOT-2017. Our model gets the second performance score (0.309) close to the best one (0.338) in terms of EAO, and with a favorable accuracy (0.50) and the second best robustness (0.27). Overall, our BCF model performs well against the state-of-the-art trackers in terms of EAO, A, and R.

C. Ablation Study

1) Effectiveness of Each Component: To verify the effectiveness of each component of the proposed method, we contrast two variants of the proposed method in different baseline frameworks (SRDCF [21] and ECO [24]): the structured GSM model (denoted by SRDCF-GSM and BCF) and the original GSM model (denoted by SRDCF-GSM-N and BCF-N). Fig. 9 shows the comparison results of these variants based on hand-crafted and deep features. We can see that with more flexibility in learning weights $\boldsymbol{\theta}$, the structured GSM model performs better than the original GSM model.

The detailed results are as follows. As shown in Fig. 9(a), compared with SRDCF-GSM-N on the original GSM model on the OTB-2015 [18] benchmark, our structured GSM model (SRDCF-GSM) gets an gain of $(0.5\%, 0.6\%)$ and $(1.1\%, 1.3\%)$ in terms of AUC and DP scores, respectively. Our SRDCF-GSM-N tracker outperforms the baseline method (SRDCF [21]) by $(1.1\%, 1.1\%)$ and $(0.2\%, 0.2\%)$ in terms of success and precision metrics, respectively. As shown in Fig. 9(b), the proposed BCF approach gets an gain of $(0.5\%, 0.5\%)$ and $(0.7\%, 0.5\%)$ for BCF-N in terms of AUC

TABLE II

PERFORMANCE EVALUATION ON VOT-2017. THE CRITERIA INCLUDE EXPECTED AVERAGE OVERLAP (EAO), ACCURACY RANK (A), AND ROBUSTNESS RANK (R). THE TOP THREE SCORES ARE HIGHLIGHTED AS RED, BLUE, AND GREEN, RESPECTIVELY

	SA-Siam	SiamRPN	GNet	MCPF	GCT	C-COT	ECO	CFCF	SiamDCF	CFWCR	CSR-DCF	MCCT	BCF	SPM
EAO	0.236	0.244	0.274	0.248	0.274	0.267	0.280	0.286	0.249	0.303	0.256	0.270	0.309	0.338
A	0.50	0.49	0.50	0.51	-	0.49	0.48	0.59	0.50	0.48	0.49	0.53	0.50	0.58
R	0.46	0.46	0.27	0.43	-	0.32	0.28	0.28	0.47	0.26	0.35	0.32	0.27	0.30

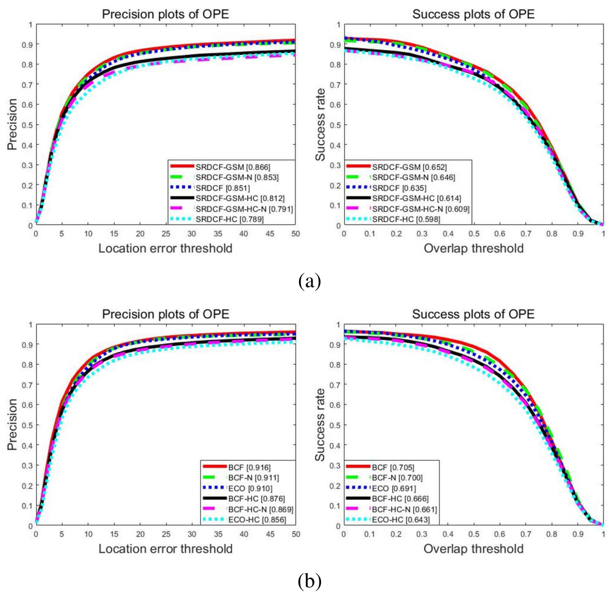

Fig. 9. The success and precision scores of each component of our model on OTB-2015 with hand-crafted features and deep features. (a) Area under curve (AUC) scores based on SRDCF, (b) Area under curve (AUC) scores based on ECO.

and DP scores, respectively. The BCF-N is still much better than the ECO model by $(1.8\%, 0.9\%)$ and $(1.3\%, 0.1\%)$ in terms of success and precision metrics, respectively. Overall, by exploiting the spatial correlations, our tracker based on the structured GSM model can achieve highly competitive results.

To better demonstrate the advantages of the structured model, we have listed six attribute-based evaluations based on the original GSM and the structured GSM model by using hand-crafted and deep features, which are shown in Fig. 10. Six attributes include: scale variation, deformation, fast motion, illumination variation, background clutter and in-plane rotation. The proposed structured GSM has performed favorably against the original GSM in those attributes. Particularly, for deformation, illumination variation, and scale variation, the structured GSM has achieved a performance improvement of $2.7\%, 1.9\%$ and 1.1% respectively over the original GSM in terms of AUC.

2) Qualitative Evaluation: We display the evaluation results of our approach with related methods on OTB-2015 benchmark, under background clutter, fast motion, occlusion, deformation, motion blur, scale variation, illumination variation and out of view. Detailed comparative results are shown in Fig. 6. For these video sequences, we can see that our model is valid for scale variation, motion blur and occlusion problems in a short time. However, some of the state-of-the-art trackers such as SRDCF [21] and ECO [24] fail.

TABLE III

COMPARISON WITH THE CLASSICAL METHODS ON OTB BENCHMARKS. THE RESULTS INCLUDE AUC SCORES AND SPEED (FPS). THE TOP THREE TRACKERS ARE SHOWN IN RED, BLUE AND GREEN, RESPECTIVELY

Tracker	AUC score (OPE)		Speed
	OTB-2013	OTB-2015	(FPS)
DSiam[2017]	0.656	-	25
SiamFC[2016]	0.607	0.582	86
BACF[2017]	0.656	0.621	35
ECO[2017]	0.709	0.691	8
C-COT[2016]	0.677	0.673	0.3
SRDCF[2015]	0.641	0.635	1
MCCT[2018]	0.714	0.696	7.8
MDNet[2016]	0.708	0.678	1
CREST[2017]	0.673	0.623	1
SPM[2019]	0.693	0.687	120
Our	0.726	0.705	3.2

3) Failure Cases: Some failure cases are shown in Fig. 11. In the first row, DRT [26] achieves better performance because its local response consistency constraint can ensure that the reliability weights will not be all concentrated around a certain area of the target. As a result, DRT can focus on more reliable parts, which makes the tracker more stable than our method. In the second and third rows, SPM [79] is able to detect targets from the background with deformations and in-plane rotation with a two-stage box refinement. These problems will be what we need to solve in the future.

To explain why the accuracy of the proposed model is lower than that of some benchmark trackers in Fig. 5 (b), we have listed the comparison results of two challenging attributes, low resolution and out-of-view, in Fig. 12. These comparison results reveal the shortcomings of the proposed model, which provide the direction for future research.

D. Complexity

To accelerate the solution of the subproblem on $\boldsymbol{h}$, we need FFT and IFFT to transform the spatial domain and frequency domain, and the corresponding complexities are $\mathcal{O}(KN)$ and $\mathcal{O}(KNlog(KN))$ for solving $\hat{\boldsymbol{h}}$ and $\boldsymbol{h}$, respectively. The complexity cost of $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$ are $\mathcal{O}(KN)$ and $\mathcal{O}(KN)$, respectively. Denote J as the iterations of alternative update for $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$, the complexity of updating $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$ is $\mathcal{O}((KN)J)$. In our experiments, good results can be achieved as $J = 1$. The complexity of element-wise operations is $\mathcal{O}(1)$. Therefore, the complexity of our optimization framework as $\mathcal{O}(KN)log(KN)I$, where I represents the number of iterations.

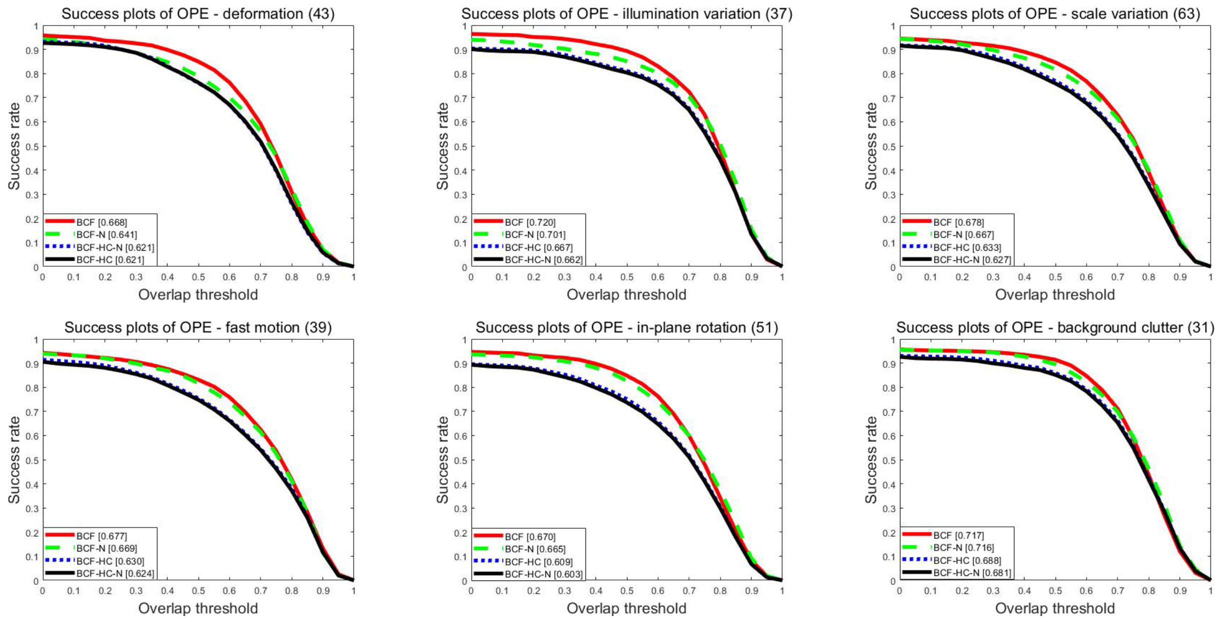

Fig. 10. The evaluation based on attributes of the OTB-2015 dataset by using hand-crafted and deep features. Here, we show the AUC scores of success plots on 6 challenges, including in-plane rotation, fast motion, deformation, scale variation, illumination variation, and background clutter. The structured GSM model outperforms the non-structured GSM model over all attributes.

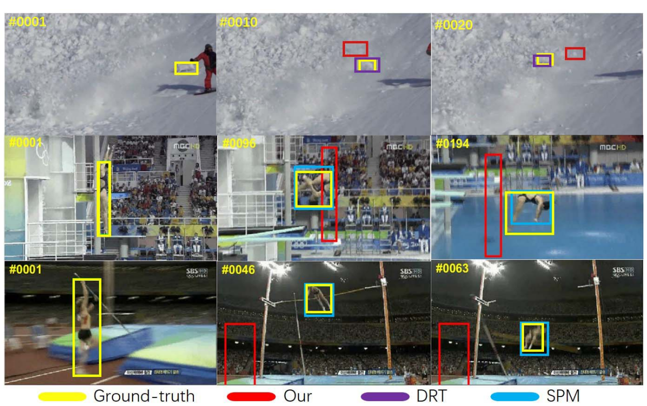

Fig. 11. Visualization of failure cases. The yellow box represents the label and the red box represents our prediction. The purple box and the blue box denote the methods of DRT and SPM, respectively.

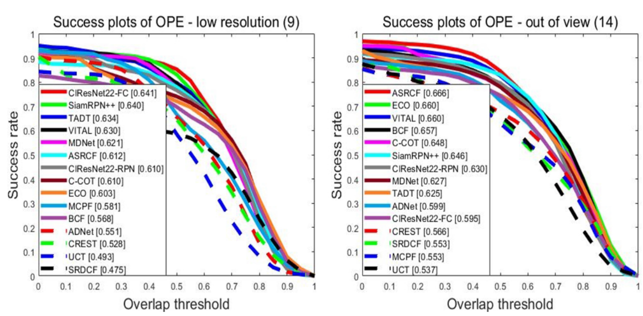

Fig. 12. The evaluation of success plots based on two challenging attributes on OTB-2015 with deep features.

The number of iterations in the optimization process not only influences the accuracy and success rates, but also affects the speed of the tracker model. We list the relationship between the number of iterations and the success rate on OTB-2015 benchmark, as well as the corresponding running speed (in terms of FPS). Detailed experimental results are shown in Fig. 13.

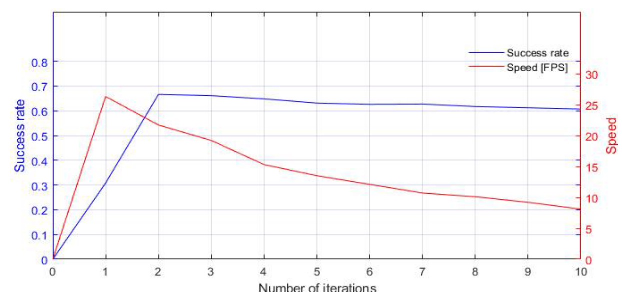

Fig. 13. The success plots and speed based on the number of iterations on OTB-2015.

Speed is also an important indicator for the evaluation of model performance. We have tested the speed of our method on the OTB benchmark, where the Frame Per Second (FPS) metric can get 21.7 FPS with hand-craft features measured on a single CPU, and the GPU version (only used in deep feature extraction) of our method operates at the speed of 3.2 FPS. Table III shows the comparison with several classical trackers, including SRDCF [21], ECO [24], C-COT [25], MDNet [29], CREST [32], MCCT [34], BACF [58], Siamfc [68], SPM [79] and DSiam [88]. Our tracking model shows improved tracking performance (in terms of AUC scores) at the sacrifice of tracking speed (in terms of FPS).

VI. Conclusion

In this paper, we proposed a novel optimization-based model to suppress the boundary effects in CF-based visual tracking. When compared with existing DCF-based trackers, our model can better handle various appearance variation-related uncertainty factors from the perspective of maximum a posterior (MAP). By modeling the spare parameters of CFs with Gaussian Scale Mixture (GSM) model, the filter can adaptively select the most representative parameters for target object localization, which strikes an improved trade-off between discrimination and reliability. Moreover, by considering the relationship of neighboring pixels of a target, we have extended the GSM model to a structured GSM model in which the accuracy of position estimation and the success rate of target detection can be further enhanced. The results from five public benchmarks demonstrate that our tracker performs favorably than most of state-of-the-art methods. In view of some weakness of the proposed model, our tracker can adopt temporal constraints in [64] or attention mechanism as mentioned in [90] for robust visual tracking, which will be the direction of our future research.

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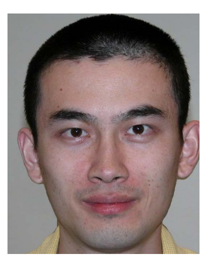

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